Base Rate Neglect Clinic Classroom Run Kit

TOK Cognitive Science Activity Bank • Mathematics • 30-40 min

Category	Mathematics	**Time**	30-40 min
Difficulty	Medium	**ID**	base-rate-neglect-clinic

Big idea: Probability judgements depend on the background rate as well as the new evidence.

One-Minute Teacher Brief

Students diagnose why a vivid test result can mislead when the underlying base rate is ignored.

Student Prompt

Inspect Base Rate Neglect Clinic Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Base Rate Neglect Clinic Stimulus A	Student-facing stimulus 1: In 1000 people, 10 have Condition X. The test catches 9 of them but falsely flags 99 people without it. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Base Rate Neglect Clinic Stimulus B	Student-facing stimulus 2: Question: if someone tests positive, is it closer to 9%, 50%, or 90% likely they have Condition X?. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Base Rate Neglect Clinic Reveal / Twist	Student-facing stimulus 3: Question: if someone tests positive, is it closer to 9%, 50%, or 90% likely they have Condition X?. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Probability judgements depend on the background rate as well as the new evidence.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Base Rate Neglect Clinic Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Base Rate Neglect Clinic Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.

#	Question for students	Teacher key / cue
6	How does representation affect mathematical knowledge?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how probability judgements depend on the background rate as well as the new evidence.

• Strong response: separates observation from inference.

• Strong response: names a method, source, model, language choice, context, or perspective.

• Strong response: revises confidence when new evidence or a new criterion appears.

• Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

• Work silently for the first response so you can notice your own starting point.

• Record one knowledge claim, the evidence behind it, and your confidence level.

• After the reveal or comparison, update your claim in a different colour or column.

• Use the debrief to answer: How does representation affect mathematical knowledge?

Setup Checklist

• Use natural frequencies rather than formulas at first.

• Keep the scenario fictional and low-stakes while preserving the structure of diagnostic reasoning.

Example Stimuli or Materials

• In 1000 people, 10 have Condition X. The test catches 9 of them but falsely flags 99 people without it.

• Question: if someone tests positive, is it closer to 9%, 50%, or 90% likely they have Condition X?

Resources To Use

• Resource pack: resources/base-rate-neglect-clinic/index.html

• Card deck CSV: resources/base-rate-neglect-clinic/cards.csv

• Visual prompt: resources/base-rate-neglect-clinic/visual-prompt.svg

• Thousand-person grid or token set with true positives and false positives in different colors.

• Risk communication card: percentage version, frequency version, and visual-grid version.

Run Sequence

1. Present a fictional condition with a low base rate and a fairly accurate test.
2. Ask students to estimate the chance someone has the condition after a positive result.
3. Build the scenario with a 1000-person or 100-person frequency table.
4. Compare intuitive estimates with the frequency result.
5. Debrief why base rates are easy to neglect.

Debrief Moves

• Knowledge question: Why can mathematically correct evidence still mislead intuitive knowers?

• When should probability be communicated as percentages and when as natural frequencies?

• How does format affect mathematical understanding?

Teacher Quick Script

• The test sounds accurate, but we need to ask: accurate among how many real cases?

• The denominator is doing quiet but essential knowledge work.

Exit Ticket

• One thing I first treated as evidence was...

• My confidence changed because...

• A better knowledge question I can now ask is...